

Lepton sector phenomenology of 1+2IDM model with fermionic triplet.

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I. INTRODUCTION

Our case does not have the problem of Strumia Minimal Dark Matter: model and results paper thanks to the Z_4 symmetry that prevents couplings with SM particles

II. THE MODEL

We consider an extended 1 + 2 Higgs doublet model, where the SM scalar sector is augmented by the inclusion of a two electrically neutral scalar singlets and the SM fermion is enlarged by a $SU(2)$ fermionic triplet. The SM gauge symmetry is enlarged by the inclusion of the discrete Z_4 symmetry, assumed to be spontaneously broken down to a preserved Z_2 symmetry, which allows for stable dark matter candidates as well as for a one loop level radiative seesaw mechanism that yields the tiny masses of active neutrinos. The scalar and leptonic particle spectrum of the model and their transformations under the $SU(3)_C \times SU(2)_L \times U(1)_Y \times Z_4$ group are shown in Tables I and II, respectively.

The full $SU(3)_C \times SU(2)_L \times U(1)_Y \times Z_4$ symmetry features the following spontaneous symmetry breaking pattern:

$$\begin{aligned} \mathcal{G} &= SU(3)_C \times SU(2)_L \times U(1)_Y \times Z_4 \\ &\quad \Downarrow v_\sigma \\ &SU(3)_C \times SU(2)_L \times U(1)_Y \times Z_2 \\ &\quad \Downarrow v \\ &SU(3)_C \otimes U(1)_Q \times Z_2 \end{aligned} \tag{1}$$

where the preserved Z_2 symmetry arises from the spontaneous breaking of the Z_4 group triggered by the vacuum expectation value of the gauge singlet scalar σ .

The scalar fields can be written as:

$$\begin{aligned} \phi &= \begin{pmatrix} \phi^+ \\ \frac{1}{\sqrt{2}}(v + \phi_R^0 + i\phi_I^0) \end{pmatrix}, & \eta_k &= \begin{pmatrix} \eta^+ \\ \frac{1}{\sqrt{2}}(\eta_{kR}^0 + i\eta_{kI}^0) \end{pmatrix}, \\ \sigma &= S + v_\sigma, & \varphi &= \frac{\varphi_R + i\varphi_I}{\sqrt{2}}, & k &= 1, 2. \end{aligned} \tag{2}$$

The leptonic multiplets can be expressed as follows:

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	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$	Z_4
ϕ	1	2	$\frac{1}{2}$	0
η_k	1	2	$\frac{1}{2}$	-1
φ	1	1	0	1
σ	1	1	0	-2

Table I: Scalar assignments under $SU(3)_C \times SU(2)_L \times U(1)_Y \times Z_4$. Here $k = 1, 2$.

	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$	Z_4
l_{iL}	1	2	$-\frac{1}{2}$	0
e_{iR}	1	1	-1	0
Ξ_R	1	3	0	-1

Table II: Lepton assignments under $SU(3)_C \times SU(2)_L \times U(1)_Y \times Z_4$.

$$l_{iL} = \begin{pmatrix} \nu_{iL} \\ e_{iL} \end{pmatrix}, \quad \Xi_R = \begin{pmatrix} \Xi^0 & \Xi^+ \\ \frac{\Xi^0}{\sqrt{2}} & -\frac{\Xi^0}{\sqrt{2}} \end{pmatrix}_R, \quad i = 1, 2, 3.$$

where l_{iL} are the SM $SU(2)_L$ lepton doublets, whereas the SM right handed charged leptons e_{iR} ($i = 1, 2, 3$) are $SU(2)_L$ singlets as usual.

With the previously specified particle content, the following leptonic Yukawa terms arise:

$$-\mathcal{L}_Y^{(l)} = \sum_{i=1}^3 \sum_{j=1}^3 (y_e)_{ij} \bar{l}_{iL} \phi e_{jR} + \sum_{i=1}^3 \sum_{n=1}^2 (y_\Xi)_{in} \bar{l}_{iL} \tilde{\eta}_n \Xi_R + z_\Xi \text{Tr} \left(\Xi_R \overline{\Xi_R^C} \right) \sigma + \text{H.c.}$$

III. SCALAR POTENTIAL

In this section we analyze the scalar sector of the model. After the spontaneous breaking of Z_4 into the preserved Z_2 symmetry, the scalar potential determines the visible CP-even spectrum, the inert charged and neutral scalar masses, and the CP-even/CP-odd mass splitting required for the one-loop generation of active neutrino masses. The most general renormalizable scalar potential invariant can be written as

$$\begin{aligned} V = & -\mu_\phi^2(\phi^\dagger\phi) - \frac{1}{2}\mu_\sigma^2\sigma^2 + \sum_{k=1}^2 \mu_{\eta_k}^2(\eta_k^\dagger\eta_k) + \left[\mu_{12}^2(\eta_1^\dagger\eta_2) + \text{H.c.} \right] + \mu_\varphi^2(\varphi^*\varphi) + \sum_{k=1}^2 \left(A_k \phi \eta_k^\dagger \varphi + \text{H.c.} \right) \\ & + \left(\frac{C}{\sqrt{2}} \varphi^2 \sigma + \text{H.c.} \right) + \lambda_\phi(\phi^\dagger\phi)^2 + \frac{1}{4}\lambda_\sigma\sigma^4 + \sum_{k=1}^2 \lambda_{\eta_k}(\eta_k^\dagger\eta_k)^2 + \frac{1}{2}\lambda_{\phi\sigma}(\phi^\dagger\phi)\sigma^2 + \gamma_1(\eta_1^\dagger\eta_1)(\eta_2^\dagger\eta_2) \\ & + \gamma_2(\eta_1^\dagger\eta_2)(\eta_2^\dagger\eta_1) + \left[\gamma_3(\eta_1^\dagger\eta_1)(\eta_1^\dagger\eta_2) + \text{H.c.} \right] + \left[\gamma_4(\eta_2^\dagger\eta_2)(\eta_2^\dagger\eta_1) + \text{H.c.} \right] + \sum_{k=1}^2 \lambda_{\eta_k\varphi}(\eta_k^\dagger\eta_k)(\varphi^*\varphi) \\ & + \left[\lambda_{12}^{(\varphi)}(\eta_1^\dagger\eta_2)(\varphi^*\varphi) + \text{H.c.} \right] + \sum_{k=1}^2 \lambda_{\eta_k\phi}(\eta_k^\dagger\eta_k)(\phi^\dagger\phi) + \left[\lambda_{12}^{(\phi)}(\eta_1^\dagger\eta_2)(\phi^\dagger\phi) + \text{H.c.} \right] \\ & + \frac{1}{2} \sum_{k=1}^2 \lambda_{\eta_k\sigma}(\eta_k^\dagger\eta_k)\sigma^2 + \frac{1}{2} \left[\lambda_{12}^{(\sigma)}(\eta_1^\dagger\eta_2)\sigma^2 + \text{H.c.} \right], \end{aligned} \quad (3)$$

where the parameters A_k and C play a distinguished role. The trilinear couplings A_k induce the mixing between the inert doublets and the inert singlet after electroweak symmetry breaking, whereas the C term splits the CP-even

and CP-odd singlet components after $\mathbb{Z}_4$ breaking. Together, these effects prevent the exact cancellation between the CP-even and CP-odd loop functions in the radiative neutrino-mass matrix.

Furthermore, the minimization conditions of the scalar potential yield the following relation

$$\begin{aligned}\mu_\phi^2 &= \lambda_\phi v^2 + \frac{1}{2}\lambda_{\phi\sigma}v_\sigma^2, \\ \mu_\sigma^2 &= \lambda_\sigma v_\sigma^2 + \frac{1}{2}\lambda_{\phi\sigma}v^2.\end{aligned}\tag{4}$$

After spontaneous symmetry breaking, the CP-even visible scalar spectrum is obtained from the diagonalization of the squared mass matrix. We find that this matrix is given by

$$\mathcal{M}_{\phi\sigma}^2 = \begin{pmatrix} M_{\phi\phi}^2 & M_{\phi\sigma}^2 \\ M_{\phi\sigma}^2 & M_{\sigma\sigma}^2 \end{pmatrix} = \begin{pmatrix} 2\lambda_\phi v^2 & \lambda_{\phi\sigma}vv_\sigma \\ \lambda_{\phi\sigma}vv_\sigma & 2\lambda_\sigma v_\sigma^2 \end{pmatrix}.\tag{5}$$

The physical CP-even visible states h_1 and h_2 are related to the original fields ϕ_R and S through the rotation angle α as

$$\begin{pmatrix} h \\ H \end{pmatrix} = R(\alpha) \begin{pmatrix} \phi_R \\ S \end{pmatrix}, \quad R(\alpha) = \begin{pmatrix} c_\alpha & s_\alpha \\ -s_\alpha & c_\alpha \end{pmatrix},\tag{6}$$

with $c_\alpha \equiv \cos \alpha$ and $s_\alpha \equiv \sin \alpha$. The mixing angle is determined by the scalar potential parameters according to

$$\tan 2\alpha = \frac{2M_{\phi\sigma}^2}{M_{\phi\phi}^2 - M_{\sigma\sigma}^2} = \frac{\lambda_{\phi\sigma}vv_\sigma}{\lambda_\phi v^2 - \lambda_\sigma v_\sigma^2}.\tag{7}$$

The corresponding eigenvalues are

$$m_{h,H}^2 = \frac{1}{2} \left[M_{\phi\phi}^2 + M_{\sigma\sigma}^2 \mp \sqrt{(M_{\phi\phi}^2 - M_{\sigma\sigma}^2)^2 + 4(M_{\phi\sigma}^2)^2} \right],\tag{8}$$

where, in the small-mixing regime, the lighter state h is identified with the SM-like Higgs boson, while H is mostly singlet-like. The mixing angle α controls the deviations of the Higgs couplings from their SM values and also enters the tree-level contribution to Direct Detection DM.

The charged inert spectrum follows from the squared mass matrix

$$\mathcal{M}_{\text{charged}}^2 = \begin{pmatrix} M_{+11}^2 & M_{+12}^2 \\ M_{+12}^2 & M_{+22}^2 \end{pmatrix},\tag{9}$$

where

$$\begin{aligned}M_{+11}^2 &= \mu_{\eta_1}^2 + \frac{1}{2}(v^2\lambda_{\eta_1\phi} + v_\sigma^2\lambda_{\eta_1\sigma}), \\ M_{+22}^2 &= \mu_{\eta_2}^2 + \frac{1}{2}(v^2\lambda_{\eta_2\phi} + v_\sigma^2\lambda_{\eta_2\sigma}), \\ M_{+12}^2 &= \mu_{12}^2 + \frac{1}{2}(v^2\lambda_{12}^{(\phi)} + v_\sigma^2\lambda_{12}^{(\sigma)}).\end{aligned}\tag{10}$$

The physical charged states are defined through

$$\begin{pmatrix} H_1^+ \\ H_2^+ \end{pmatrix} = R(\theta) \begin{pmatrix} \eta_1^+ \\ \eta_2^+ \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{pmatrix},\tag{11}$$

with $c_+ \equiv \cos \theta_+$ and $s_+ \equiv \sin \theta_+$. The mixing angle satisfies

$$\tan 2\theta = \frac{2M_{+12}^2}{M_{+11}^2 - M_{+22}^2} = \frac{2v^2\lambda_{12}^{(\phi)} + 2v_\sigma^2\lambda_{12}^{(\sigma)} + 4\mu_{12}^2}{v^2(\lambda_{\eta_1\phi} - \lambda_{\eta_2\phi}) + v_\sigma^2(\lambda_{\eta_1\sigma} - \lambda_{\eta_2\sigma}) + 2(\mu_{\eta_1}^2 - \mu_{\eta_2}^2)}.\tag{12}$$

The charged scalar masses are

$$m_{H_{1,2}^+}^2 = \frac{1}{2} \left[M_{+11}^2 + M_{+22}^2 \mp \sqrt{(M_{+11}^2 - M_{+22}^2)^2 + 4(M_{+12}^2)^2} \right]. \quad (13)$$

The neutral inert sector splits into CP-odd and CP-even blocks. In the basis $(\eta_{1I}, \eta_{2I}, \varphi_I)$, the CP-odd squared mass matrix is

$$\mathcal{M}_{\text{odd}}^2 = \begin{pmatrix} M_{I11}^2 & M_{I12}^2 & -\frac{v}{\sqrt{2}}A_1 \\ M_{I12}^2 & M_{I22}^2 & -\frac{v}{\sqrt{2}}A_2 \\ -\frac{v}{\sqrt{2}}A_1 & -\frac{v}{\sqrt{2}}A_2 & \mu_\varphi^2 - 2Cv_\sigma \end{pmatrix}, \quad (14)$$

where

$$\begin{aligned} M_{I11}^2 &= \mu_{\eta_1}^2 + \frac{1}{2} (v^2 \lambda_{\eta_1 \phi} + v_\sigma^2 \lambda_{\eta_1 \sigma}), \\ M_{I22}^2 &= \mu_{\eta_2}^2 + \frac{1}{2} (v^2 \lambda_{\eta_2 \phi} + v_\sigma^2 \lambda_{\eta_2 \sigma}), \\ M_{I12}^2 &= \mu_{12}^2 + \frac{1}{2} (v^2 \lambda_{12}^{(\phi)} + v_\sigma^2 \lambda_{12}^{(\sigma)}). \end{aligned} \quad (15)$$

In the CP-even basis $(\eta_{1R}, \eta_{2R}, \varphi_R)$, one obtains

$$\mathcal{M}_{\text{even}}^2 = \begin{pmatrix} M_{R11}^2 & M_{R12}^2 & \frac{v}{\sqrt{2}}A_1 \\ M_{R12}^2 & M_{R22}^2 & \frac{v}{\sqrt{2}}A_2 \\ \frac{v}{\sqrt{2}}A_1 & \frac{v}{\sqrt{2}}A_2 & \mu_\varphi^2 + \sqrt{2}Cv_\sigma \end{pmatrix}, \quad (16)$$

with

$$M_{Rij}^2 = M_{Iij}^2, \quad i, j = 1, 2. \quad (17)$$

Using Eqs. (7) and (8) we can define

$$\begin{aligned} R(\alpha)^T \begin{pmatrix} m_{h_1}^2 & 0 \\ 0 & m_{h_2}^2 \end{pmatrix} R(\alpha) &= \begin{pmatrix} c_\alpha & -s_\alpha \\ s_\alpha & c_\alpha \end{pmatrix} \begin{pmatrix} m_{h_1}^2 & 0 \\ 0 & m_{h_2}^2 \end{pmatrix} \begin{pmatrix} c_\alpha & s_\alpha \\ -s_\alpha & c_\alpha \end{pmatrix} \\ &= \begin{pmatrix} c_\alpha^2 m_{h_1}^2 + s_\alpha^2 m_{h_2}^2 & s_\alpha c_\alpha (m_{h_1}^2 - m_{h_2}^2) \\ s_\alpha c_\alpha (m_{h_1}^2 - m_{h_2}^2) & c_\alpha^2 m_{h_2}^2 + s_\alpha^2 m_{h_1}^2 \end{pmatrix} \\ &\Downarrow \\ \lambda_{\phi\sigma} v v_\sigma &= s_\alpha c_\alpha (m_h^2 - m_H^2) \end{aligned} \quad (18)$$

$$\lambda_{\phi\sigma} = 0.87 \quad \alpha = 0.579 \quad (19)$$

Tree level boundeness from below conditions Accidental symmetry.

Collider constraints for charged scalars in [1]. The charged inert scalars $H_k^\pm$ arise from two $SU(2)_L$ doublets with hypercharge $Y = 1/2$. Therefore, their electroweak quantum numbers are $Q = 1$ and $t_3^{(+1)} = 1/2$. Their pair production through the Drell-Yan process, $pp \rightarrow H_k^+ H_k^-$, is thus fixed by their masses, $M_{H_k^\pm}$ and electroweak quantum numbers, and is independent of the charged-scalar mixing angle θ_+ . The relevant physical constraint we must impose is $m_{\Xi^0} < M_{H_k^\pm}$.

IV. NEUTRINO MASSES.

The masses of active neutrinos arise at one-loop level. The neutrino mass matrix has the following form:

$$(M_\nu)_{ij} = \sum_{n,r=1}^2 \sum_{k=1}^3 \frac{(y_\Xi)_{in} (y_\Xi)_{jr} m_\Xi}{16\pi^2} \left[(R_{\chi_R})_{nk} (R_{\chi_R})_{rk} \frac{m_{\chi_{kR}}^2}{m_{\chi_{kR}}^2 - m_\Xi^2} \ln \left(\frac{m_{\chi_{kR}}^2}{m_\Xi^2} \right) - (R_{\chi_I})_{nk} (R_{\chi_I})_{rk} \frac{m_{\chi_{kI}}^2}{m_{\chi_{kI}}^2 - m_\Xi^2} \ln \left(\frac{m_{\chi_{kI}}^2}{m_\Xi^2} \right) \right], \quad (20)$$

where

$$\begin{pmatrix} \eta_{1R} \\ \eta_{2R} \\ \varphi_R \end{pmatrix} = R_{\chi_R} \begin{pmatrix} \chi_{1R} \\ \chi_{2R} \\ \chi_{3R} \end{pmatrix}, \quad \begin{pmatrix} \eta_{1I} \\ \eta_{2I} \\ \varphi_I \end{pmatrix} = R_{\chi_I} \begin{pmatrix} \chi_{1I} \\ \chi_{2I} \\ \chi_{3I} \end{pmatrix},$$

are the rotation matrices of the CP-even and CP-odd scalar states, respectively. In order to fit this model to experimental data, we employ the Casas-Ibarra parametrization. Defining the matrix Λ as

$$\Lambda_{nr} = \sum_{k=1}^3 \frac{m_\Xi}{16\pi^2} \left[(R_{\chi_R})_{nk} (R_{\chi_R})_{rk} \frac{m_{\chi_{kR}}^2}{m_{\chi_{kR}}^2 - m_\Xi^2} \ln \left(\frac{m_{\chi_{kR}}^2}{m_\Xi^2} \right) - (R_{\chi_I})_{nk} (R_{\chi_I})_{rk} \frac{m_{\chi_{kI}}^2}{m_{\chi_{kI}}^2 - m_\Xi^2} \ln \left(\frac{m_{\chi_{kI}}^2}{m_\Xi^2} \right) \right],$$

we can express the mass matrix for the active neutrinos as follows

$$(M_\nu)_{ij} = \sum_{n=1}^2 \sum_{r=1}^2 (y_\Xi)_{in} \Lambda_{nr} (y_\Xi)_{jr} = \sum_{n=1}^2 \sum_{r=1}^2 (y_\Xi)_{in} \Lambda_{nr} (y_\Xi^T)_{rj}.$$

$$M_\nu = y_\Xi \Lambda y_\Xi^T = y_\Xi R_\Lambda \Lambda_d R_\Lambda^T y_\Xi^T = \tilde{y}_\Xi \Lambda_d \tilde{y}_\Xi^T, \quad R_\Lambda^T \Lambda R_\Lambda = \Lambda_d, \quad y_\Xi = \tilde{y}_\Xi R_\Lambda^T$$

this allows the Yukawa coupling matrix to be expressed as

$$y_\Xi = U^* \sqrt{m_\nu} R^T \sqrt{\Lambda_d^{-1}} R_\Lambda,$$

where

$$m_\nu = \begin{pmatrix} 0 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix} \quad (21)$$

for neutrinos in normal hierarchy and U is the experimental PMNS matrix.

$$M_\nu = U^* \sqrt{m_\nu} R \sqrt{\Lambda_d^{-1}} \Lambda_d \sqrt{\Lambda_d^{-1}} R^T \sqrt{m_\nu} (U^*)^T = U^* m_\nu (U^*)^T,$$

$$U^T M_\nu U = m_\nu.$$

Using this parametrization, both the mass of the triplet Ξ and the angle ϕ are treated as input parameters. By requiring the resulting values of Δm_{21}^2 , Δm_{32}^2 , $\sin^2 \theta_{12}^{(\ell)}$, $\sin^2 \theta_{23}^{(\ell)}$, $\sin^2 \theta_{13}^{(\ell)}$, and $\delta_{CP}^{(\ell)}$ to lie within the experimental values at 3σ , we can perform a scan in order to obtain the matrix element $|y_\Xi| < \sqrt{4\pi}$.

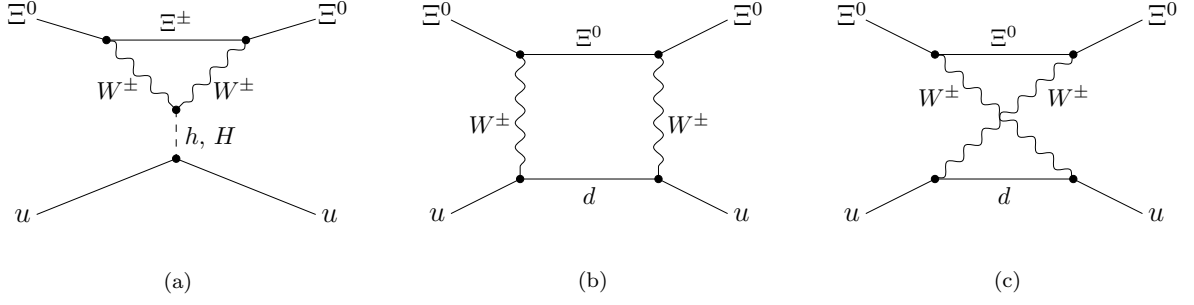


Figure 1: Pure 1-loop EW contributions to elastic nucleon-DM scattering

V. FERMIONIC DARK MATTER.

A. Relic density

The main contribution to fermionic triplet annihilation comes from co-annihilation between neutral and charged states of the triplet, product of the radiative mass splitting between neutral and charged components ($\Delta m_\Xi \sim 166$ MeV). In usual perturbative tree-level computations, the experimental relic abundance can be reproduced by a fermionic triplet with mass around $m_\Xi \simeq 2.3$ TeV [2]. Non-perturbative effects like Sommerfeld enhancement can strongly modify the prediction of the relic abundance, moving to masses up to $m_\Xi \simeq 3$ TeV [3]. Here, we adopt the value $m_\Xi = 2.6$ TeV for the mass reported in Ref. [4].

B. Direct detection

Unlike smaller isospin multiplets, the main contribution to elastic scattering of non-hypercharged fermionic triplet with nucleons comes from the one-loop electroweak mediated Feynman diagrams of Fig. 1 as was discussed in Ref. [5]. Robust treatments (for example in Ref. [6]) show that, for triplet scenario, the predictions of direct detection are very close to the current experimental limit of LZ [7]. This requires special attention in model-dependent contributions like tree-level or extra one-loop contributions and the interference effects with the leading EW one.

The relevant effective Lagrangian can be written as [7, 8]

$$\mathcal{L}_{\text{eff}}^{\text{SI}} = f_q \mathcal{O}_S^q + f_G \mathcal{O}_S^G + \sum_{i=q,G} \left[\frac{g_1^i}{m_\Xi} \mathcal{O}_{T_1}^i + \frac{g_2^i}{m_\Xi^2} \mathcal{O}_{T_2}^i \right], \quad (22)$$

where the scalar operators are

$$\mathcal{O}_S^q = (\bar{\Xi}\Xi) m_q \bar{q}q, \quad \mathcal{O}_S^G = (\bar{\Xi}\Xi) \frac{\alpha_s}{\pi} G_{\mu\nu}^a G^{a\mu\nu}, \quad (23)$$

and the twist-2 operators are

$$\mathcal{O}_{T_1}^i = i \bar{\Xi} \partial^\mu \gamma^\nu \Xi \mathcal{O}_{\mu\nu}^i, \quad \mathcal{O}_{T_2}^i = - \bar{\Xi} \partial^\mu \partial^\nu \Xi \mathcal{O}_{\mu\nu}^i. \quad (24)$$

For the quark and gluon energy-momentum tensor operators we use

$$\mathcal{O}_{\mu\nu}^q = \frac{i}{2} \bar{q} \left(D_\mu \gamma_\nu + D_\nu \gamma_\mu - \frac{1}{2} g_{\mu\nu} \not{D} \right) q, \quad \mathcal{O}_{\mu\nu}^G = G_\mu^{a,\rho} G_{\nu\rho}^a - \frac{1}{4} g_{\mu\nu} G_{\rho\sigma}^a G^{a,\rho\sigma}. \quad (25)$$

The pure electroweak contribution is matched at the electroweak scale $\mu_Z \simeq m_Z$. The relevant scalar Wilson coeffi-

cients are collected as

$$\vec{C}_5^{\text{EW}}(\mu_Z) = \begin{pmatrix} f_u^{\text{EW}} \\ f_d^{\text{EW}} \\ f_s^{\text{EW}} \\ f_c^{\text{EW}} \\ f_b^{\text{EW}} \\ f_G^{\text{EW}} \end{pmatrix}_{\mu_Z}. \quad (26)$$

For the real triplet, the light-quark coefficients satisfy

$$f_u^{\text{EW}} = f_d^{\text{EW}} = f_s^{\text{EW}} = f_c^{\text{EW}} \equiv f_q^{\text{EW}}, \quad (27)$$

whereas f_b^{EW} will be treated separately. For numerical values we use the $N_f = 2 + 1$ lattice scheme. The explicit matching coefficients for f_q^{EW} , f_b^{EW} , f_G^{EW} , and for the twist-2 Wilson coefficients $g_{1,2}^{q,G}$, are given in Appendix.

In addition to this, the model contains the scalar-mediated tree-level contribution in Figure ???. The relevant interactions in the physical scalar basis are

$$\mathcal{L} \supset -z_\Xi (s_\alpha h + c_\alpha H) \bar{\Xi}\Xi - \sum_q \frac{m_q}{v} (c_\alpha h - s_\alpha H) \bar{q}q, \quad (28)$$

Integrating h and H gives

$$\mathcal{L}_{\text{eff}}^{\text{tree}} = f_q^{\text{tree}} \sum_q \text{Tr}(\bar{\Xi}\Xi) m_q \bar{q}q, \quad (29)$$

with

$$f_q^{\text{tree}} = \frac{z_\Xi}{v} s_\alpha c_\alpha \left(\frac{1}{m_h^2} - \frac{1}{m_H^2} \right) = \frac{m_\Xi}{4vv_\sigma} \sin 2\alpha \left(\frac{1}{m_h^2} - \frac{1}{m_H^2} \right) \quad (30)$$

using the mass relation implied by the singlet vacuum expectation value $z_\Xi = m_\Xi/(2v_\sigma)$. We implement the scalar-mediated effect as a shift of the five-flavour Wilson coefficients at the electroweak matching scale. In the convention $\mathcal{O}_S^G = (\bar{\Xi}\Xi)(\alpha_s/\pi)G_{\mu\nu}^a G^{a\mu\nu}$, the induced gluonic coefficient is approximated by the standard Higgs-mediated matching relation $f_G^{\text{tree}} = -\frac{1}{8}f_q^{\text{tree}}$ **we need to clarify the -1/8 factor**. Thus, the tree-level scalar exchange induces the shift $f_q^{\text{EW}} \rightarrow f_q^{\text{EW}} + f_q^{\text{tree}}$ and $f_G^{\text{EW}} \rightarrow f_G^{\text{EW}} - \frac{1}{8}f_q^{\text{tree}}$. The scalar coefficient vector used as the initial condition for the numerical running is

$$\vec{C}_5^{\text{tot}}(\mu_Z) = \vec{C}_5^{\text{EW}}(\mu_Z) + \Delta \vec{C}_5^{\text{tree}}(\mu_Z) = \begin{pmatrix} f_q^{\text{EW}} + f_q^{\text{tree}} \\ f_q^{\text{EW}} + f_q^{\text{tree}} \\ f_q^{\text{EW}} + f_q^{\text{tree}} \\ f_q^{\text{EW}} + f_q^{\text{tree}} \\ f_b^{\text{EW}} + f_q^{\text{tree}} \\ f_G^{\text{EW}} - \frac{1}{8}f_q^{\text{tree}} \end{pmatrix}_{\mu_Z}. \quad (31)$$

The scalar Wilson coefficients are then evolved to the hadronic scale μ_{had} using the QCD running and heavy-quark threshold matching matrices summarized in Appendix ???. Explicitly,

$$\vec{C}_3^{\text{tot}}(\mu_{\text{had}}) = R_3(\mu_{\text{had}}, \mu_c) M_3(\mu_c) R_4(\mu_c, \mu_b) M_4(\mu_b) R_5(\mu_b, \mu_Z) \vec{C}_5^{\text{tot}}(\mu_Z). \quad (32)$$

We denote the resulting low-energy scalar coefficients by

$$\vec{C}_3^{\text{tot}}(\mu_{\text{had}}) = \begin{pmatrix} f_u^{\text{tot}} \\ f_d^{\text{tot}} \\ f_s^{\text{tot}} \\ f_G^{\text{tot}} \end{pmatrix}_{\mu_{\text{had}}}. \quad (33)$$

The scalar contribution to the nucleon amplitude is then

$$k_S^{\text{tot}} = \sum_{q=u,d,s} f_q^{\text{tot}}(\mu_{\text{had}}) f_{Tq}^{(N)} + f_G^{\text{tot}}(\mu_{\text{had}}) f_{TG}^{(N)}. \quad (34)$$

The gluonic matrix element $f_{TG}^{(N)}$ is reconstructed from the QCD trace anomaly using the same perturbative truncation employed in the running of the scalar Wilson coefficients.

The twist-2 contribution is not modified by the scalar exchange at tree level. It is therefore kept equal to the pure electroweak result,

$$k_T^{\text{EW}} = \frac{3}{4} \sum_{q=u,d,s,c,b} [q(2, \mu_Z) + \bar{q}(2, \mu_Z)] [g_1^q(\mu_Z) + g_2^q(\mu_Z)] - \frac{3}{4} g(2, \mu_Z) [g_1^G(\mu_Z) + g_2^G(\mu_Z)], \quad (35)$$

where $q(2, \mu_Z)$, $\bar{q}(2, \mu_Z)$, and $g(2, \mu_Z)$ are the second moments of the quark, antiquark and gluon PDFs evaluated at the electroweak scale.

The full spin-independent amplitude at the nucleon level is therefore

$$\mathcal{A}_N^{\text{tot}} = m_N (k_S^{\text{tot}} + k_T^{\text{EW}}). \quad (36)$$

Finally, the spin-independent elastic cross section per nucleon is

$$\sigma_{\text{SI}}^N = \frac{4}{\pi} \mu_{\Xi N}^2 |\mathcal{A}_N^{\text{tot}}|^2, \quad \mu_{\Xi N} = \frac{m_{\Xi} m_N}{m_{\Xi} + m_N}. \quad (37)$$

In the heavy-DM limit, $\mu_{\Xi N} \simeq m_N$, this reduces to

$$\sigma_{\text{SI}}^N \simeq \frac{4}{\pi} m_N^4 |k_S^{\text{tot}} + k_T^{\text{EW}}|^2. \quad (38)$$

$$\frac{\alpha_s}{12\pi v} G_{\mu\nu} G^{\mu\nu} (\cos \alpha h - \sin \alpha H) \quad (39)$$

C. Results

The accepted points are obtained from a global fit including neutrino oscillation observables and Higgs-sector observables. In particular, the global chi-square includes

$$\Delta m_{21}^2, \quad \Delta m_{32}^2, \quad \sin^2 \theta_{12}, \quad \sin^2 \theta_{13}, \quad \sin^2 \theta_{23}, \quad \delta_{\text{CP}}, \quad R_{\gamma\gamma}, \quad \kappa_V, \quad \kappa_F. \quad (40)$$

We retain points satisfying

$$\chi_{\text{tot}}^2 < 9, \quad (41)$$

which is used as a practical criterion to identify points simultaneously compatible with neutrino data and precision Higgs measurements.

We also require that the scale spontaneous breaking of the $\mathbb{Z}_4 \rightarrow \mathbb{Z}_2$ discrete symmetry is not below 10 TeV, ensuring that the breaking occurs above enough the mass scale of the fermionic triplet.

To characterize the interference between the two amplitudes, we define the ratio

$$r_N \equiv \frac{\mathcal{A}_N^{\text{tree}}}{\mathcal{A}_N^{\text{EW}}}. \quad (42)$$

It then follows that

$$\frac{\sigma_{\text{SI}}^N}{\sigma_{N,\text{EW}}^N} = |1 + r_N|^2. \quad (43)$$

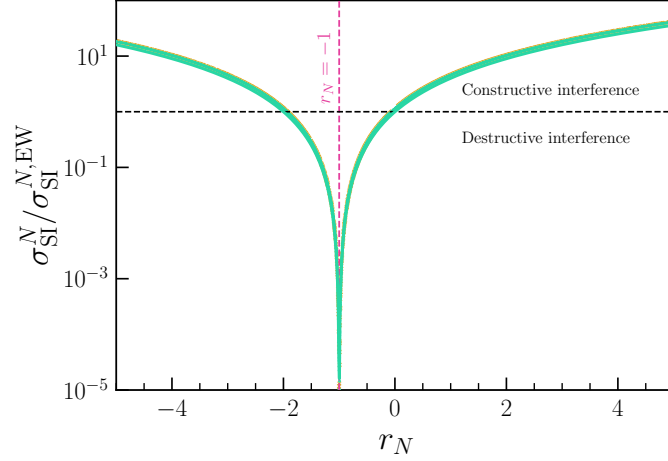


Figure 2: Spin-independent cross section normalized to the purely electroweak prediction as a function of the amplitude ratio r_N . The region of destructive interference is localized near $r_N \simeq -1$.

This relation shows that the scattering rate is suppressed when r_N approaches -1 , namely when the tree-level scalar amplitude interferes destructively with the electroweak one-loop amplitude. This behaviour is displayed in Fig. 2. The scan points follow the expected amplitude-level interference pattern.

The tree-level scalar contribution is controlled by the mixing between the active doublet and the visible singlet. It is useful to introduce the dimensionless effective parameter

$$\zeta_{\Xi}^{\text{DD}} \equiv \frac{m_{\Xi}}{v_{\sigma}} \sin 2\alpha \left(1 - \frac{m_h^2}{m_H^2}\right) = 2z_{\Xi} \sin 2\alpha \left(1 - \frac{m_h^2}{m_H^2}\right), \quad (44)$$

such that $f_q^{\text{tree}} = \zeta_{\Xi}^{\text{DD}} / (4vm_h^2)$. Thus, for $z_{\Xi} > 0$ and $m_H > m_h$, the branch $\zeta_{\Xi}^{\text{DD}} < 0$ corresponds to $\sin 2\alpha < 0$ and is the only branch that can generate destructive interference with the electroweak amplitude.

The size of ζ_{Ξ}^{DD} required for destructive interference can be estimated from the electroweak amplitude. The purely electroweak spin-independent cross section is of order $\sigma_{\text{SI}}^{N,\text{EW}} \sim 10^{-47} \text{ cm}^2$, which corresponds to a nucleon-level amplitude of order

$$|\mathcal{A}_N^{\text{EW}}| \sim 10^{-10} \text{ GeV}^{-2}. \quad (45)$$

Using

$$\frac{1}{4vm_h^2} \simeq 6.5 \times 10^{-8} \text{ GeV}^{-3}, \quad (46)$$

one finds that values in the range

$$|\zeta_{\Xi}^{\text{DD}}| \sim 10^{-3} - 10^{-2} \quad (47)$$

generate scalar coefficients of order

$$|f_q^{\text{tree}}| \sim 10^{-10} - 10^{-9} \text{ GeV}^{-3}. \quad (48)$$

After including the gluon matching and the scalar hadronic matrix elements, this scale induces a nucleon-level amplitude of the same order as $\mathcal{A}_N^{\text{EW}}$.

Figure 3 shows this correlation in the numerical scan. The condition $r_N \simeq -1$ selects a narrow band in the negative branch $\zeta_{\Xi}^{\text{DD}} < 0$.

Separating the positive and negative branches of ζ_{Ξ}^{DD} makes the role of the sign of the scalar contribution explicit. As shown in Fig. 4, for $\zeta_{\Xi}^{\text{DD}} < 0$ the cross section develops a valley associated with destructive interference. For too small values of $|\zeta_{\Xi}^{\text{DD}}|$, the scalar amplitude is subdominant and the purely electroweak prediction is recovered. For too large values, the tree-level contribution dominates the cross section again. In contrast, for $\zeta_{\Xi}^{\text{DD}} > 0$ the interference is constructive and the cross section increases with the size of the effective scalar contribution.

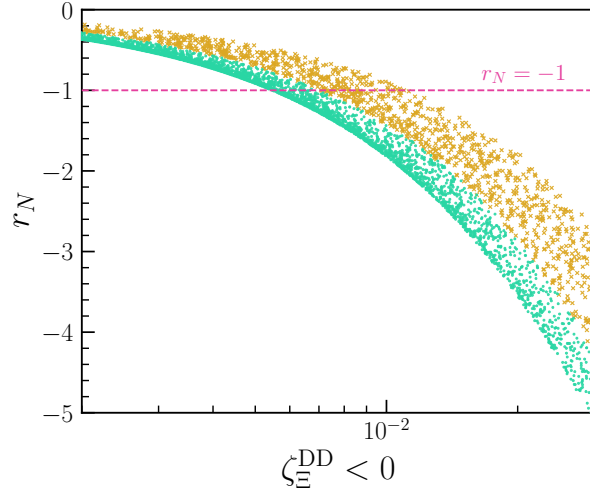


Figure 3: Amplitude ratio r_N as a function of the effective variable ζ_{Ξ}^{DD} in the negative branch. Maximal destructive interference is obtained when $r_N \simeq -1$, selecting a band with $|\zeta_{\Xi}^{\text{DD}}|$ of order a few 10^{-3} .

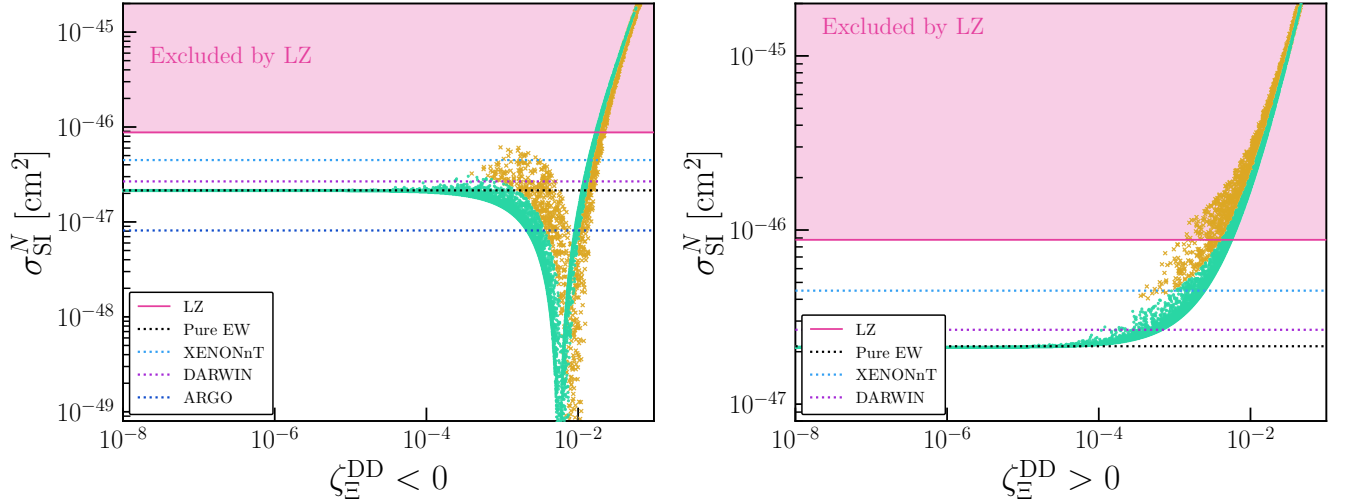


Figure 4: Spin-independent cross section as a function of the effective variable ζ_{Ξ}^{DD} , separating the negative and positive branches. The negative branch contains the region of destructive interference, whereas the positive branch only produces constructive interference. The horizontal line indicates the purely electroweak contribution, and the shaded regions show current limits and projected sensitivities from direct detection experiments.

Finally, Fig. 5 shows the total direct-detection prediction as a function of the heavy CP-even scalar mass. The reference line corresponds to the purely electroweak contribution of the fermionic triplet, while the spread of points around this line is generated by the tree-level scalar contribution. Points above the electroweak prediction correspond to constructive interference or to a scalar-dominated regime. Points below it correspond to the destructive-interference band. Therefore, the total prediction can lie above, close to, or below the purely electroweak contribution depending on the sign and magnitude of ζ_{Ξ}^{DD} .

The main result is that the direct-detection prediction of the fermionic triplet is not fixed solely by the electroweak one-loop contribution. The visible scalar sector can significantly modify it through amplitude-level interference. However, a relevant suppression of σ_{SI}^N requires both the correct sign and an appropriate magnitude of the effective variable ζ_{Ξ}^{DD} . The destructive-interference region is therefore localized in a negative band of ζ_{Ξ}^{DD} of order a few 10^{-3} . It does not represent a generic suppression over the full parameter space, but rather a specific parametric condition involving z_{Ξ} , α , and m_H .

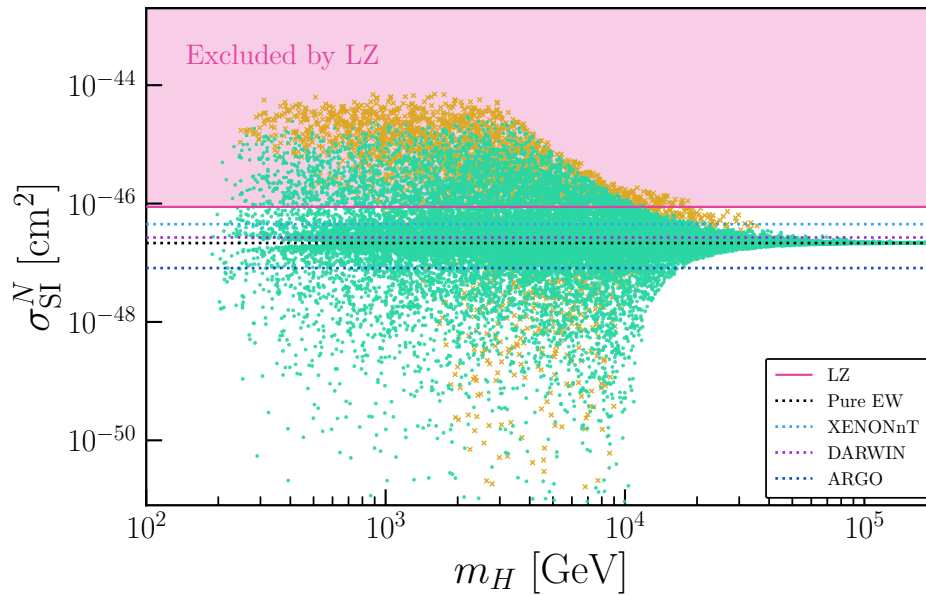


Figure 5: Total spin-independent cross section as a function of the heavy CP-even scalar mass m_H . The black dotted line indicates the purely electroweak prediction of the fermionic triplet. The experimental bands show the current LZ limit and projected sensitivities of future experiments. The spread of points reflects the interference between the electroweak one-loop amplitude and the tree-level scalar amplitude.

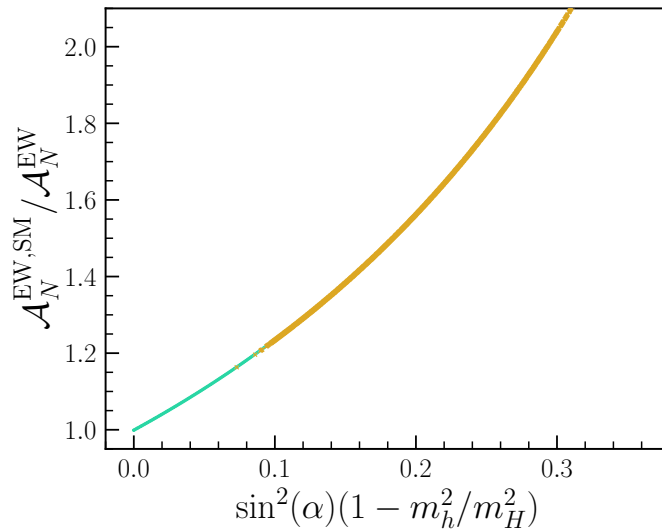


Figure 6

D. DM Indirect Detection.

E. Collider signals for the fermionic triplet.

We make qualitative comments on collider signals for the fermionic triplet.

VI. CHARGED LEPTON FLAVOR VIOLATION.

$$Br(l_i \rightarrow l_j \gamma) = \frac{3(4\pi)^3 \alpha_{\text{em}}}{4G_F^2} \left| A_D^{(i,j)} \right|^2 Br(l_i \rightarrow l_j \nu_i \bar{\nu}_j), \quad (49)$$

$$A_D^{(i,j)} = \sum_{k=1}^2 \sum_{s=1}^2 \frac{(x_{\Xi})_{is} (x_{\Xi})_{js}}{2(4\pi)^2 m_{\eta_k^\pm}^2} F\left(\frac{m_{\Xi_R}^2}{m_{\eta_k^\pm}^2}\right) \quad (50)$$

$$F(x) = \frac{1 - 6x + 3x^2 + 2x^3 - 6x^2 \ln x}{6(1-x)^4} \quad (51)$$

$$(x_{\Xi})_{is} = \sum_{k=1}^3 (y_{\Xi})_{ks} \left(V_{lL}^\dagger \right)_{ik} \quad (52)$$

here we are using $V_{lL} = \mathbb{I}_{3 \times 3}$.

VII. LEPTOGENESIS.

Due to scalars

VIII. CONCLUSIONS.

IX. ACKNOWLEDGMENTS

Appendix A: Scotogenic one-loop contribution

The neutral inert sector generates an additional one-loop correction to the effective $\Xi\Xi h_i$ vertices, with $h_i = h, H$, through diagrams containing an active neutrino and two CP-even or CP-odd inert scalars. Since the visible scalar couples to the external quarks through the SM Yukawa interaction, these Higgs-penguin diagrams modify only the scalar operator,

$$\mathcal{O}_S^q = (\bar{\Xi}\Xi) m_q \bar{q} q, \quad (A1)$$

and do not generate twist-2 operators. We define the physical trilinear couplings according to

$$\mathcal{L} \supset -\frac{1}{2} G_{\rho R_a R_a} \rho R_a^2 - \frac{1}{2} G_{\rho I_a I_a} \rho I_a^2, \quad \rho = h, H. \quad (A2)$$

For diagonal scalar contributions, the corresponding correction to the flavor-universal scalar Wilson coefficient is

$$f_q^{\text{scoto}} = \frac{m_{\Xi}}{64\pi^2 v} \sum_{\ell=1}^3 \sum_{a=1}^3 \left\{ - \left| \sum_{n=1}^2 (R_{\chi_R})_{na} (y_{\Xi})_{\ell n} \right|^2 F_L(M_{R_a}) \left[\frac{c_{\alpha} G_{h R_a R_a}}{m_h^2} - \frac{s_{\alpha} G_{H R_a R_a}}{m_H^2} \right] \right. \\ \left. + \left| \sum_{n=1}^2 (R_{\chi_I})_{na} (y_{\Xi})_{\ell n} \right|^2 F_L(M_{I_a}) \left[\frac{c_{\alpha} G_{h I_a I_a}}{m_h^2} - \frac{s_{\alpha} G_{H I_a I_a}}{m_H^2} \right] \right\}. \quad (A3)$$

The scalar coefficient entering the direct-detection amplitude is therefore shifted as

$$f_q^{\text{scalar}} = f_q^{\text{tree}} + f_q^{\text{scoto}}. \quad (A4)$$

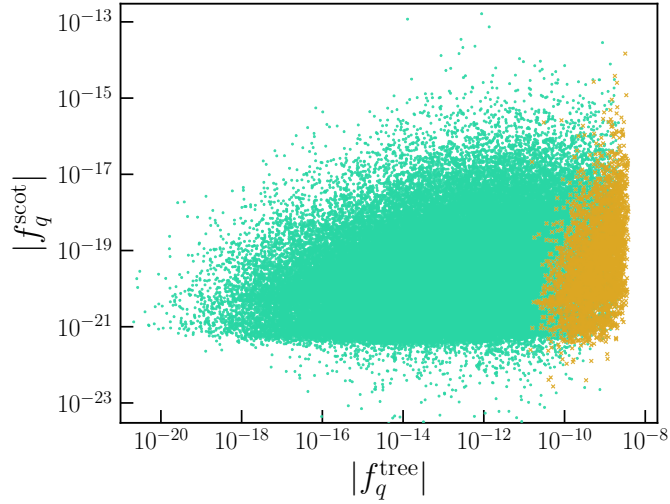


Figure 7: Scotogenic one-loop correction f_q^{scoto} as a function of the tree level Wilson coefficient f_q^{tree} .

Neglecting active-neutrino masses and taking the zero momentum-transfer limit, the exact loop function for two identical internal scalars is

$$F_L(M_X) = -\frac{1}{M_X^2} \frac{r_X + (1 - r_X) \ln(1 - r_X)}{r_X^2}, \quad r_X = \frac{m_\Xi^2}{M_X^2}. \quad (\text{A5})$$

This expression applies in the region considered in the scan, $M_X > m_\Xi$, where F_L is real. The function does not include the global loop factor already displayed in the definition of f_q^{scoto} .

Figure 7 shows that the accepted scan points satisfy

$$|f_q^{\text{scoto}}| \lesssim 10^{-15} \text{ GeV}^{-3}. \quad (\text{A6})$$

The loop correction may become comparable to f_q^{tree} in isolated regions where the tree-level contribution is itself strongly suppressed. Nevertheless, its absolute magnitude remains well below that required to modify the interference between the tree-level scalar amplitude and the electroweak contribution. The scotogenic Higgs-penguin correction can therefore be neglected in the baseline direct-detection prediction for the scanned parameter region.

Appendix B: Numerical BP

For the time being, this section will include the scalar BP for discussion.

χ^2	2.24091
Higgs-SM	125.178
CP-Even ($m_{H1}, m_{\eta_{1R}}, m_{\eta_{2R}}, m_{FR}$)	{12508.3, 7628.53, 10047.7, 10240.7}
CP-Odd ($m_{\eta_{1I}}, m_{\eta_{2I}}, M_{F1}$)	{7626.63, 10060.3, 10229.1}
Charged ($m_{\eta_{1-}}, m_{\eta_{2-}}$)	{7628.11, 10060.1}
splitting 1,2,3	{1.90052, -12.5923, 11.5685}
A1r, A2r, c1r	{7522.99, -2859.33, 8033.06}
$\lambda\phi r, \lambda\phi r, \lambda 1r, \lambda 2r, \lambda 3r, \lambda 4r, \lambda 5r, \lambda 6r, \lambda 7r, \lambda 8r, \lambda 9r,$ $\lambda 12r, \lambda 13r, \lambda 14r$	{0.57813, 0.922229, 0.857278, 1.11065, 1.2865, -1.25697, -0.814487, -1.24226, 1.22377, 1.25477, -0.729778, 1.23622, 1.31132, 1.48706}
$\mu 1r, \mu 2r, \mu 12r, \mu cr$	{4126.97, 1930.76, 101.884 - 17.4944 i, 4291.21}
$\gamma 1r, \gamma 2r, \gamma 3r, \gamma 4r$	{-0.567255, -1.36191, -0.455335 + 0.854728 i, 0.500832 + 0.373726 i}
vor, o1	{9208.46, -0.0186503}

Figure 8: Numerical fit. All masses and trilinear coupling are in GeV

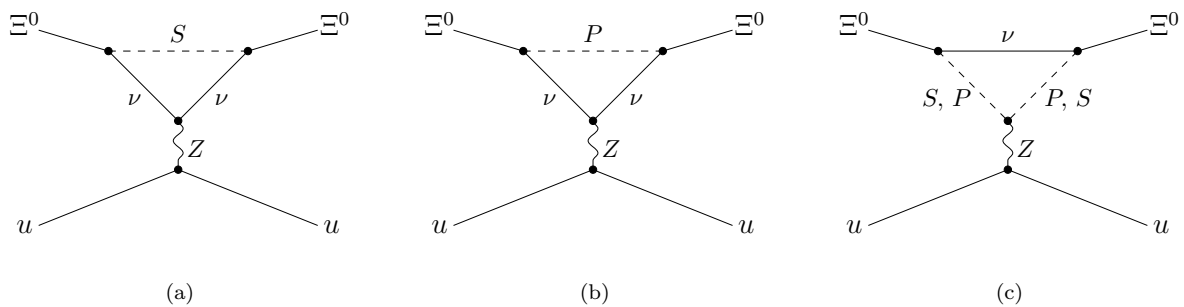


Figure 9: 1-loop relevant Scoto-EW contributions to elastic nucleon-DM scattering

Appendix C: Feynman diagrams for radiative elastic nucleon-DM scattering

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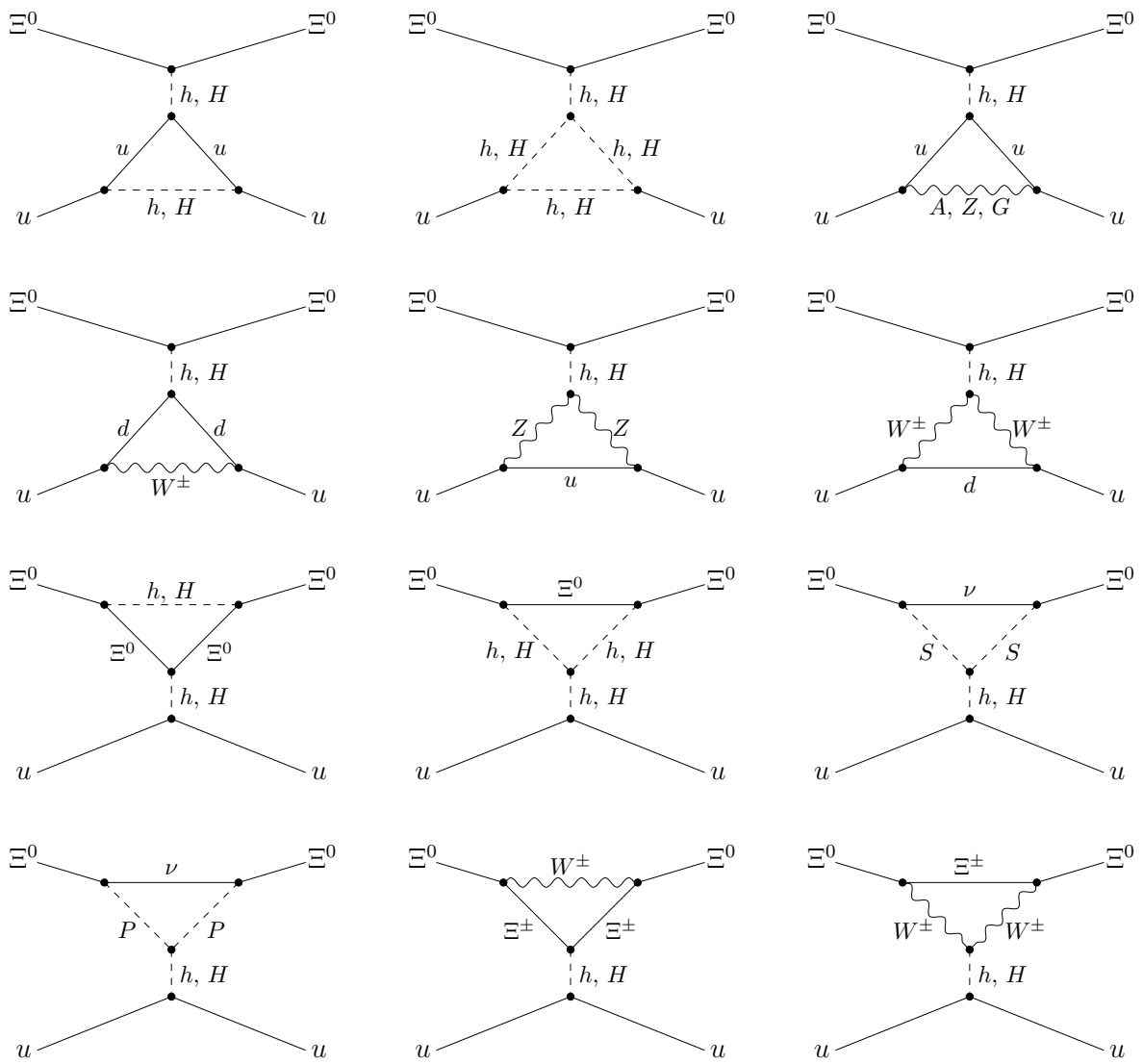


Figure 10

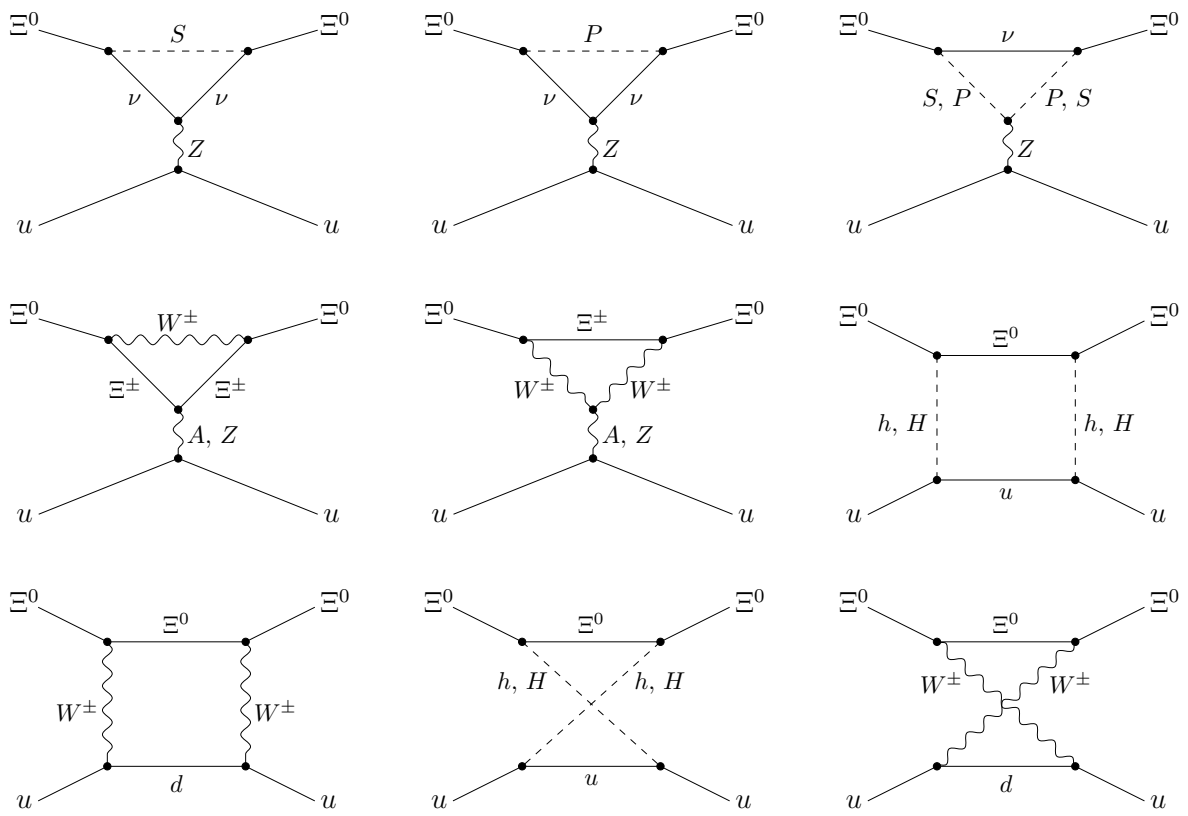


Figure 11